WORKSHEET

Chapter 9: General equation of a straight line

1 Given the following y-intercepts and gradients, write the equation of each line in the form $y = a + bx$.

a y-intercept 5, gradient 2 $y =$ _____

b y-intercept 0, gradient 3 $y =$ _____

c y-intercept 3, gradient -2 $y =$ _____

d y-intercept 2, gradient -1 $y =$ _____

e y-intercept -2, gradient $\frac{1}{2}$ $y =$ _____

f y-intercept 1, gradient -4 $y =$ _____

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g y-intercept -4, gradient 2 $y =$ _____

h y-intercept 3, gradient 0 $y =$ _____

i y-intercept -1, gradient $-\frac{3}{4}$ $y =$ _____

j y-intercept 4, gradient 1 $y =$ _____

2 Graph each of the lines described in question **1** above in the space provided.

3 Each table represents a linear function. Find the equation of each function in the form $y = a + bx$.

a

x	0	1	2	3
y	3	6	9	12

b

x	-1	0	1	2
y	7	6	5	4

c

x	0	1	2	3
y	-2	-1	0	1

d

x	-1	0	1	2
y	2	4	6	8

e

x	0	1	2	3
y	5	2	-1	-4

f

x	-1	0	1	2
y	-2	2	6	10

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g

x	2	3	4	5.5
y	5	5	5	5

h

x	4	5	6	7
y	19	24	29	34

4 Find the equation of the line that passes through the points:

a (0, 2) and (4, 10)

b (2, 5) and (3, 1)

c (1, 8) and (6, 13)

d $(-1, -3)$ and (2, 6)

e $(-3, 8)$ and $(1, -4)$

f $(-2, 3)$ and (4, 3)

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Answers

1 a $y = 5 + 2x$

b $y = 3x$

c $y = 3 - 2x$

d $y = 2 - x$

e $y = -2 + \frac{1}{2}x$

f $y = 1 - 4x$

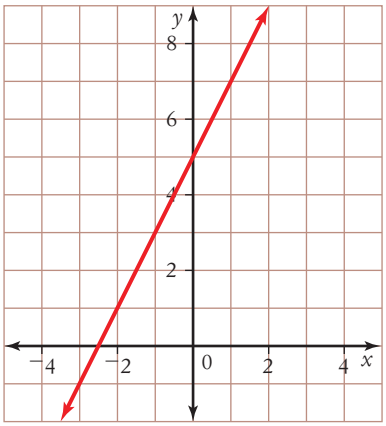
g $y = -4 + 2x$

h $y = 3$

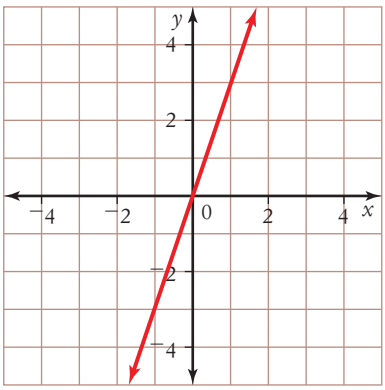
i $y = -1 - \frac{3}{4}x$

j $y = 4 + x$

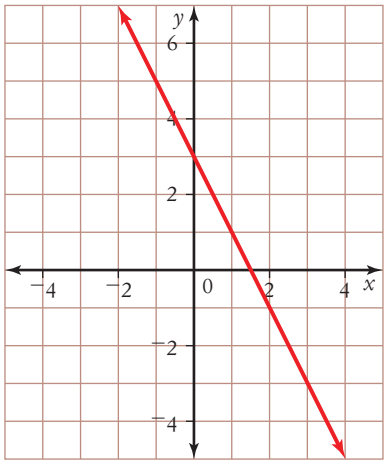
2 a



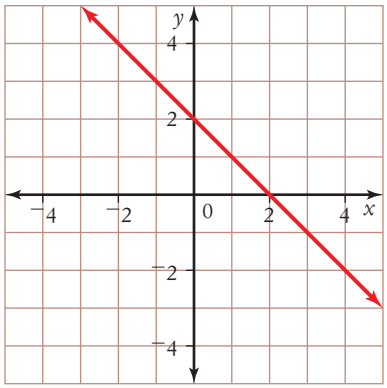
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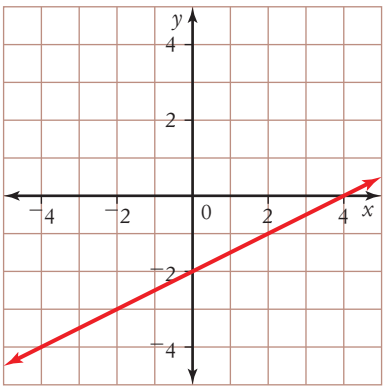
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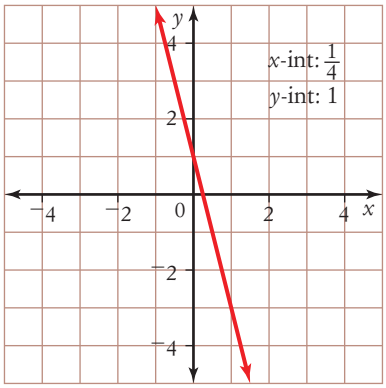


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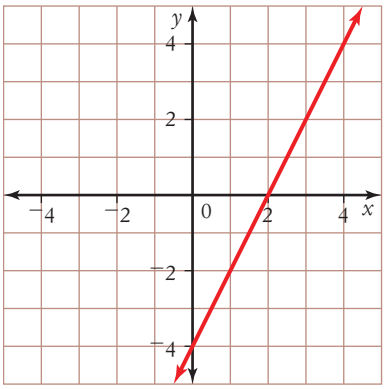


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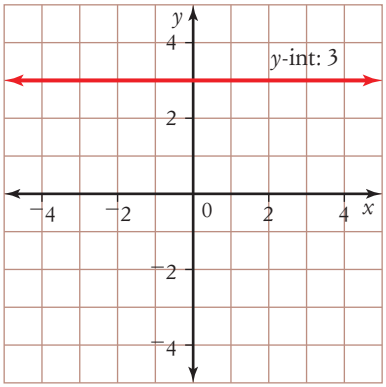
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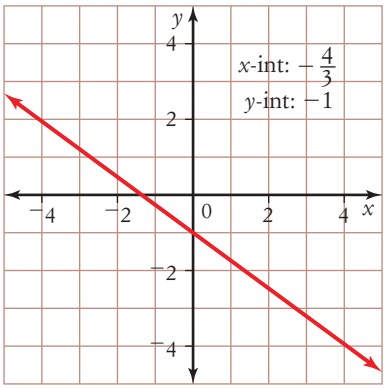
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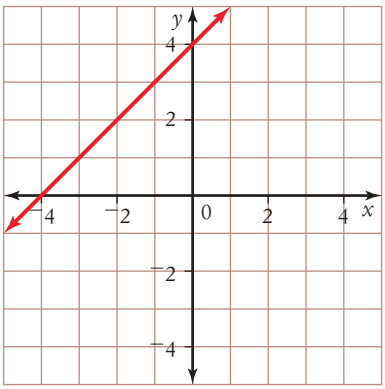
h



i



j



3 **a** $y = 3 + 3x$
b $y = 6 - x$
c $y = -2 + x$
d $y = 4 + 2x$
e $y = 5 - 3x$
f $y = 2 + 4x$
g $y = 5$
h $y = -1 + 5x$

4 **a** $y = 2 + 2x$
b $y = 13 - 4x$
c $y = 7 + x$
d $y = 3x$
e $y = -1 - 3x$
f $y = 3$

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